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Chromosome mechanics and architecture

In our lab we study the fundamental properties of DNA, the main information carrier in biology. With highly focused laser beams we can trap microscopic objects and apply forces on them. These so-called optical tweezers (Nobel Prize Physics 2018) enable us to measure the mechanical properties of DNA while we can concurrently use fluorescence microscopy to visualize the interactions of proteins with the DNA.

In cells however, DNA is not a "bare" molecule, but it is tightly compacted by a large amount of proteins into a structure called the chromosome. To transfer information from mother to daughter cells, one copy of each chromosome has to go into the two daughter cells during cell division. This is when the famous X-shaped chromosomes appear, in which the two copies of the DNA are still attached to each other at the crossing point called the centromere. There are estimates that during cell division, forces as high as 500 pN act on the centromere. In comparison, the double helical shape of bare DNA already gets lost at forces of about 65 pN. It is a key question in biology how chromosomes can withstand such high forces.

In this project, we quantify the mechanical properties of chromosomes. We collaborate with biologists who attach isolated chromosomes to microspheres (beads) that we can trap with optical tweezers and that allow accurate force measurements on chromosomes.

We are looking for someone that is highly motivated to play a key role in developing this method and performing experiments on chromosomes which may include the first direct force measurements on centromere-labeled chromosomes and 3D-STED super-resolution imaging of chromosomes. The student will learn techniques like optical tweezers, fluorescence microscopy and microfluidics and obtain hands-on experience in working with biological materials such as DNA and proteins. If you are enthusiastic about this project and about working in an international team of people with varying backgrounds, please **contact Gijs Wuite: g.j.l.wuite@vu.nl, Erwin Peterman (e.j.g.peterman@vu.nl) or Iddo heller (i.heller@vu.nl).**

Study of DNA – protein interactions using optical tweezers

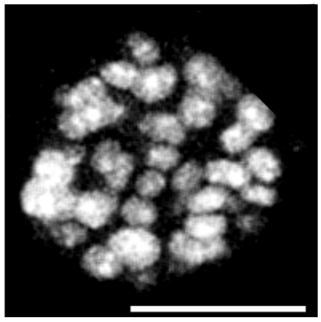
Helicases are enzymes that act as motor proteins to separate or remodel DNA or RNA. They play a key role in various cellular processes, including DNA replication, repair and recombination. RecQ helicases are one family of DNA helicases that have a highly conserved helicase domain and are required to maintain genome stability. Mutation in these helicases results in chromosomal instability and in diseases that have a predisposition to cancer. Due to their importance in living organisms, helicases are emerging as an important class of targets for antiviral, antibiotic and anti-cancer drugs.

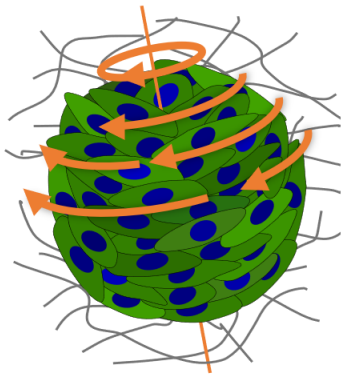
This project focuses on the helicase BLM which regulates DNA replication, recombination, and can also resolve aberrant structures formed during these processes. The molecular mechanism of BLM will be studied using a combination of optical tweezers and fluorescence microscopy which is a powerful tool used to apply precise and localized forces while simultaneously visualizing single molecules in real-time. Additionally, you will also use DNA constructs that illustrate the structure of DNA found in cells during replication and investigate the interaction of BLM with other proteins in order to understand its mechanism.

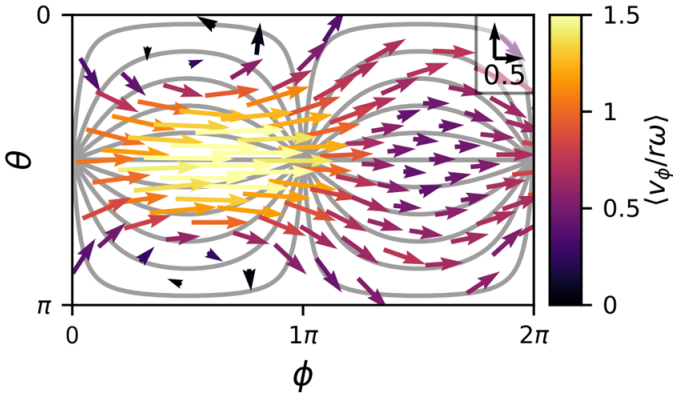
Contact Shyla Singh (s.singh@vu.nl)

Collective migration of spherically shaped cell clusters

Collective cell migration is a process where cells move in a coordinated way to accomplish biological tasks. Often, cells migrate collectively in three-dimensional curved environments. Such curved geometries arise naturally, with examples ranging from developing embryos to tumors. A major outstanding challenge for these systems, is to understand how three-dimensional collective cell migration is impacted by the system's curved geometry. In this theoretical project, we study this challenge using an *in vitro* model system known as an epithelial spheroid. These spheroids are spherically shaped cell clusters consisting of around 100 cells. Strikingly, these spheroids perform global rotations superficially resembling that of planetary rotations. However, these rotations exhibit a previously unexplored pattern in the form of a velocity wave and topological defects of coordinated cell migration. It remains challenging, however, to determine whether such patterns arise from biological mechanisms, or as a consequence of generic physical phenomena. Using living theoretical concepts like that of active matter, we aim to uncover the biophysical principles of how and why cells establish such collective patterns. In this project you will take part in our ongoing research to theoretically elucidate how collective cell migration patterns emerge in spheroids from the interplay between single-cell motility, cell-cell interactions, and the system's curved geometry.







Contact: **Chase Broedersz** (c.p.broedersz@vu.nl)

Theory of bacterial chromosome organization

In all organisms, chromosomal DNA needs to be highly compacted to fit inside its cellular confinement. Many bacteria contain only a single chromosome but with a contour length a thousand-fold larger than the cell. The necessary chromosomal compaction poses a major organizational problem: the DNA does not only have to be highly condensed to fit inside the cell, but its dynamic organization also has to facilitate crucial genetic processes. Understanding the functional organization of bacterial chromosomes in living developing cells is a key outstanding challenge. Our group is approaching this problem by using maximum entropy modelling, polymer simulations, and statistical approximations to gain insight into the dynamics of chromosome organization, higher-order chromosomal structure, and the effect of loop-extruding proteins on chromosome organization. We are currently recruiting MSc students interested in using computational and statistical approaches to study chromosome organization.

Contact: **Chase Broedersz** (c.p.broedersz@vu.nl)

Characterizing the impact of nucleotide analog Molnupiravir-TP on SARS-CoV-2 polymerase

Description

The recent outbreak of SARS-CoV-2 has infected millions of human worldwide, showing how dreadful impact these RNA-viruses can have on the society. To be prepared for any future outbreak, the search for an effective antiviral drug against SARS-CoV-2 is still ongoing. The main lacking is the poor understanding of the mode of incorporation of these nucleotides analogs that target the coronavirus polymerase to impair its essential activity.

To make new copies of the viral RNA, the SARS-CoV-2 forms replication-transcription complex (RTCs) using several non-structural proteins (NSPs). The core unit of this complex is referred as RNA-dependent RNA-polymerase (RdRp) consisting of three non-structural proteins (NSP 7, NSP 8 & NSP 12).

In our lab we have established a single molecule magnetic tweezers assay to study in vitro the RNA synthesis process by the SARS-CoV-2 RdRp [1]. Using our assay we can also study the effect of different antiviral drugs on the RdRp at a single molecule level [2]. In this project, we will characterize the impact of the nucleotide analog molnupiravir tri-phosphate on RNA synthesis by SARS-CoV-2 RdRp using our already established high-throughput magnetic tweezers assay.

We are seeking for an enthusiastic and motivated student to join this project, which offers an opportunity to gain hands-on experience in magnetic tweezers and data analysis using Python. If you are interested to work on this project, or would like to ask questions/discuss possibilities, please contact **Asif Rakib (a.rakib@vu.nl)** or **David Dulin (d.dulin@vu.nl).**

[1] Subhas Chandra Bera, Mona Seifert, Robert N. Kirchdoerfer, et al., The nucleotide addition cycle of the SARS-CoV-2 polymerase, Cell Reports, Volume 36, Issue 9, 2021.

[2] Mona Seifert, Subhas C Bera, Pauline van Nies, et al. (2021), Inhibition of SARS-CoV-2 polymerase by nucleotide analogs from a single-molecule perspective, eLife 10:e70968.

Bsc/Msc project description: Studying RNA replication/biogenesis by SARS-CoV-2

The recent COVID-19 outbreak demonstrated the devastating impact RNA viruses, in particular coronaviruses (CoVs), can have on society. In depth knowledge of the molecular processes taking place during viral infection is key to ready ourselves for a new outbreak.

Severe Acute Respiratory Syndrome coronavirus 2 (SARS-CoV-2), the causative agent of COVID-19, uses a myriad of proteins to process its RNA genome, together forming the “replication-transcription complex” (RTC). The core unit is an RNA-dependent RNA-polymerase (RdRp), capable of producing a complementary RNA from a template. While copying the RNA, the RTC uses helicases to resolve secondary structures, methyl- and guanine- transferases to chemically edit the product, and an exonuclease to remove wrongly incorporated nucleotides.

How do these proteins come together to form an RTC? How does the RTC coordinate the activity of its different enzymes, if at all? How does the RTC respond to antivirals?

These are some of the questions we will attempt to answer using a combination of single-molecule magnetic tweezers, fluorescence, and mathematical modeling.

Contact: **Misha Klein** (m.klein2@vu.nl) or **David Dulin (d.dulin@vu.nl).**

Investigating the Impact of NSP9 on SARS-CoV-2 Replication Transcription Complex (RTC) through Magnetic Tweezers

The outbreak of COVID-19 caused by the SARS-CoV-2 virus has spurred significant scientific interest in understanding its replication mechanisms to develop effective therapeutic interventions. Among the viral proteins involved, NSP9 has emerged as a potential target due to its role in viral replication and transcription. This project aims to investigate the effect of NSP9 on the SARS-CoV-2 Replication Transcription Complex (RTC) by analyzing dsRNA constructs using magnetic tweezers, a powerful technique for studying biophysical properties at the molecular level.

The findings could have implications for the development of antiviral strategies targeting NSP9 or other components of the RTC, thereby aiding in the design of more effective therapies against COVID-19.

We are seeking an enthusiastic and motivated student to join this project, which offers an opportunity to gain hands-on experience in performing experiments with magnetic tweezers and data analysis using Python routine. If you are interested to work on this project, or would like to ask questions/discuss possibilities, please contact Arnab Das (a.das2@vu.nl) or David Dulin (d.dulin@vu.nl).

Relevant literature:

** Subhas Chandra Bera, Mona Seifert et al, The nucleotide addition cycle of the SARS-CoV-2 polymerase, Cell Reports, 2021
** Nora Schmidt et al, SND1 binds SARS-CoV-2 negative-sense RNA and promotes viral RNA synthesis through NSP9, Cell, 2023

E. coli RNAP devoid of the ω subunit and its susceptibility to external agents

In bacteria, transcription of the DNA into messenger RNA (mRNA) is carried out by the multi-subunit RNA polymerase (RNAP), which comprises core and accessory subunits. One of these, the ω subunit, was overlooked for long time and the first bulk studies were suggesting minor relevance compared to the other subunits of the core RNAP.

Later studies have demonstrated its chaperone activity on the β' subunit, known to load different factors regulating transcription initiation (Bhowik *et al.*, 2017) and elongation (Sarkar *et al.*, 2013), and its involvement in the response to stress signals (Vrentas *et al.*, 2005). The presence of the ω subunit further regulates transcription in a fashion that is related to the topology of the DNA (Geertz *et al.*, 2011), that can be more relaxed or more compact, the latter in the case of supercoiling.

Through an established high-throughput magnetic tweezers assay, we are able to study bacterial TX on a single-molecule level, allowing investigation on how the dynamics of the process are affected and shed light how other factors can be dysregulated by its absence.

Contact: **Luca Buccolieri** (l.buccolieri@vu.nl) or **David Dulin (d.dulin@vu.nl).**

Single-molecule investigation of antibiotics-related oxidative stress

An oxygen-rich environment and an aerobic metabolism for energy production come with the drawback of oxidative stress, disrupting of the internal homeostasis of the system (Kumar *et al.*, 2004). Antibiotics are an exogenous cause of oxidative stress in bacteria, altering the building blocks of life. One of the most relevant products is 8-oxo-guanosine (8-oxo-G), incorporated in DNA as 8-oxo-dG, causing mutagenesis, or in mRNA as is, increasing aberrant proteins' production (Agapov *et al.*, 2019). Altogether, these processes further compromise the bacterial cell, increasing antibiotics efficacy.

Through an established high-throughput magnetic tweezers assay, we are able to study bacterial mRNA synthesis (transcription, TX) on a single-molecule level, hence the possibility of investigating how its impacted by these modifications and how different TX factors may intervene (Agapov *et al.*, 2019) or be affected (Kumar *et al.*, 2004) by these conditions.

Contact: **Luca Buccolieri** (l.buccolieri@vu.nl) or **David Dulin (d.dulin@vu.nl).**

A BIOPHYSICAL APPROACH TO STUDY THE MAMMALIAN INNATE IMMUNITY:

In our lab, we focus on researching how the innate immune system responds to viral infections, using a biophysical approach. In mammalian cells, a complex yet tightly regulated cascade of events occur during viral infections. These events are triggered by Pathogen Recognition Receptors (PRR), leading to an effective immune response.

We particularly focus on the Melanoma differentiation associated gene 5 (MDA5), which is one of the first PRR's that detects viral RNA in the cell. Upon recognition, this protein forms a filament with viral RNA and interacts with Mitochondrial Anti-Viral Signaling (MAVS) proteins present on the surface of mitochondria. This in turn triggers the production of interferons, which can result in the death of viral infected cells.

However, several aspects of MDA5 functionality are currently unknown or unclear. In addition to already establishing how this protein changes the mechanical properties of dsRNA, we would like to further research how it distinguishes between mammalian RNA and viral RNA, the dynamics of filament assembly on dsRNA and how environmental conditions such as pH may influence MDA5 recognition of dsRNA. Moreover, how certain derived peptides may inhibit the function of MDA5, which may be useful in treating interferonopathies.
Also, how certain dsRNA modifications may alter the detection by MDA5. This is important to study as these modifications may be used by viruses to escape immune detection. To approach this, we will employ the use of single molecule force spectroscopy techniques such as magnetic tweezers to understand the dynamics involved.

We are looking for a motivated student who is interested in learning these force spectroscopy techniques to understand the role of MDA5 from a single molecule perspective. This internship will also provide you with an opportunity to learn data analysis using Python.

For further details or internship opportunities, please contact Zainab Rashid (z.m.rashid@vu.nl) or David Dulin (d.dulin@vu.nl).

Physics of antifreeze proteins and ice crystal growth

Growth of ice crystals can induce critical damage to soft condensed matter systems such as living tissue and cells. Many lifeforms produce ice-binding proteins (IBPs) for protection against frost-damage at temperatures as low as -70C. It is yet unclear how IBPs can function as such remarkable cryoprotectants. Although models exist that relate IBP activity to pinning of advancing ice planes, numerous open questions regarding the microscopic mechanisms remain. We search for answers to these mechanistic questions by studying ice crystal growth and probing IBP activity at the molecular length scales where IBPs act. To this end, we are developing innovative single-molecule methodology based on optical tweezers, (super-resolution) fluorescence microscopy and novel (laser) refrigeration methods to directly resolve dynamic interactions between ice, water, and proteins on the nanometer length scale. Projects are available on all aspects of this research, ranging from theoretical modeling, new methods development, to biophysical experimentation. Examples include:

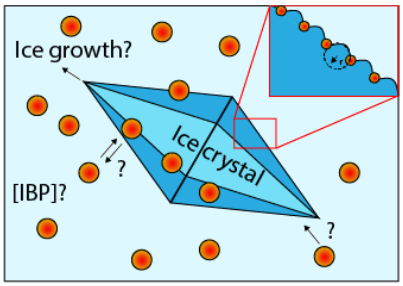
- Develop thermo-electric cooling in optical tweezers/fluorescence microscopy systems.
- Explore fluorescence polarization anisotropy to quantify local temperature profiles in optical tweezers.
- Experimental study of ice growth and ice recrystallization (Ostwald Ripening) and the effect of ice binding proteins.
- Perform computational and theoretical modeling of ice crystal growth and the impact of ice binding proteins.

Contact: Iddo Heller (i.heller@vu.nl) *(see also separate info docs on physics of ice binding proteins and laser refrigeration in optical tweezers **below**)*

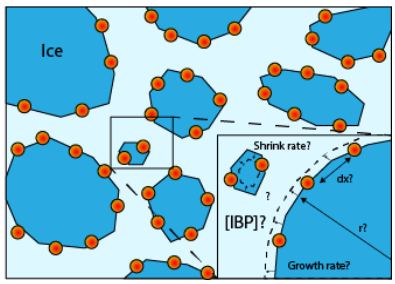
The physics of antifreeze proteins

Ice binding proteins can be found in a wide variety of living systems, and act as a cryo-protective agent for internal frost damage. By adhesion to the ice surface, these proteins are able to modulate the crystallization process is a non-colligative way. Protein activity can be divided into two categories (see below). We aim to elucidate and possibly unify the biophysical principles that underlie IBP activity. Two major IBP functions can be distinguished:

- **Thermal Hysteresis (TH)**: in freeze avoiding species, IBPs arrest growth of embryonic ice crystals through the Thermal Hysteresis effect, which is a phenomenon of lowering the freezing point of a solution and simultaneous increase of the melting point. In between adsorption sites, ice supposedly grows in a convex fashion, increasing surface energy and making further ice growth energetically unfavorable. Protein activity is an interplay between solvent concentration adsorption rates, surface coverage, ice growth and plane affinity, but the exact mechanism is still unknown.



- **Ice Recrystallization Inhibition (IRI)**: in freeze tolerant species, large volume fractions of ice are not necessarily fatal to the organisms, but IBPs are required to keep the average ice crystal size small through **Ice Recrystallization Inhibition (IRI)**. Natural recrystallization, known as Ostwald ripening, is a diffusion driven process minimizing surface free energy by growing bigger crystals at the expense of smaller ones. IBP's at a critical concentration can transition this diffusive-driven process to a reaction-driven process, inhibiting the recrystallization process and stabilizing the system at an average crystal radius.



Ice Recrystallization Inhibition

Intriguingly, all known IBPs capable of causing thermal hysteresis effect also show IRI activity, already at ~1000 times lower concentrations. Although this suggests a shared physical origin, it remains unclear how these two strategies correlate.

Within this theme 3 projects are available:

1) **Elucidate the physics of IBP activity through theoretical and computational modelling**. In particular, we aim to explore a possible role of the local curvature effect (Gibbs-Thomson effect) and kinetic pinning in ice recrystallization inhibition in order to investigate the link between the two roles IBPs perform. See: *Impurity blocking of growth steps: experiments and theory,* van Enckevort et al., 1996
2) **Quantify the effect of IBP binding at a single-molecule level.** Fluorescently labelled proteins can be localized on a crystal with high resolution using confocal microscopy. With this approach we want to understand the ice binding process and the local effect that single protein can have on a crystal region or binding plane. Using different variants of the same IBP (with and without the TH activity) we aim to visualize differences in binding patterns.
3) **Develop image processing workflow for quantitative analysis of crystal shape evolution.** The fluorescence images of IBPs provide insight into the binding mechanism of the proteins, but only the brightfield recordings of growing crystals allow full interpretation of the physical conditions during imaging. The goal of the project is to establish workflow for improved data collection and time efficient analysis of the two datasets.

Contact: Iddo Heller (i.heller@vu.nl), Thomas Reus (t.j.n.reus@vu.nl)

Exploring giant unilamellar vesicle mechanics with OTs (BSc project)

Giant unilamellar vesicles (GUVs) are cell-sized membrane-only structures, often used as model systems to investigate the mechanobiology of more complex biological samples.

In this project, you will learn how to produce GUVs via an inverted emulsion technique, and use single-molecule techniques to characterise their mechanics. Furthermore, by introducing polymers inside the liposomes (e.g. agarose or collagen), we also want to test how internal components affect the force response. In order to do this, you will be using double-trap optical tweezers to perform pulling experiments on GUVs; by fluorescently labelling the membrane lipids, you will be able to track membrane deformations over time.

In order to analyse this type of data, you will be working on a contour detection algorithm (Python or ImageJ) to obtain membrane shape changes as a function of time/force.

With this project, we aim to assess how individual components, such as membrane or an internal 'skeleton', are involved in the response to force application. This is particularly important in the context of artificial cells, as the latter require a certain mechanical resilience to be fully functional.

Contacts: Andreas Biebricher (a.s.biebricher@vu.nl), Gijs Wuite (g.j.l.wuite@vu.nl)

Student project available in Erwin Peterman's lab

Website: https://www.nat.vu.nl/~erwinp/
Contact: e.j.g.peterman@vu.nl
Summary of the work done in video format (recommended to watch):
https://www.youtube.com/watch?v=T_ZjuqJLTr0&ab_channel=WormLabVrijeUniversiteit

Using advanced fluorescence microscopy to study transport and sensing in living, multicellular organisms

In our lab, we focus on elucidating the dynamics of intracellular transport in the chemosensory cilia in the nematode ('worm') *Caenorhabditis elegans* and its role in the sensing of chemicals (like salt, repellants and toxic components: 'taste'). To this end we use advanced single-molecule fluorescence microscopy, in combination with image analysis, data analysis and simulations. We generate mutant worm strains that express fluorescent versions of proteins involved in transport and sensing. In our microscopy experiments, we use tricks like photoactivation, selective photobleaching, microfluidics. Internships are available focusing on all different aspects of the projects: from instrumentation development, to development of data/image analysis tools, to applying these tools to an interesting worm strain, to the generation of new worm strains using crosses or CRISPR/Cas9.

Current projects:

In this project you will study chemosensation: the process of detecting the chemical environment and responding to it. To study this process we make use of the nematode (worm) *C. elegans*. When a worm is crawling forward and comes in contact with a repellent (a chemical compound that it does not like) it shows avoidance behaviour: it crawls backwards and then turns around to change direction. How this process takes place on a molecular and cellular level is not entirely understood. We have therefore made a microfluidics device with which we can stimulate the worm with a chemical solution while simultaneously imaging the neuronal activity.